Table 1: Free and open source simulators for IoT integrated smart city implementations

Simulator/framework	Features
CupCarbon	Smart city simulation, 2D-3D OpenStreetMap visualisation, SCI-WSN simulator, Google Map Views, satellite views; *http://www.cupcarbon.com*
Contiki	Smart city and industrial simulation, enabled with IPv6, IPv4, low resource, microcontrollers, proto-threading, game consoles; *http://www.contiki-os.org*
InterSCity	Smart city simulation, trips definition and processing, city routing, city visualisation; *http://interscity.org*
OpenIoT	Smart city simulation, actuators and smart sensors, Sensing-as-a-Service (S2aaS); *http://www.openiot.eu*
Zetta	WebSocket (low overhead, real-time on TCP), reactive programming; *http://www.zettajs.org*
SCSimulator	Smart cities, city resource distribution, traffic management; *https://github.com/ezambomsantana/smart-city-simulator*
DSA	Inter-device communication, logic and apps on all layers; *http://www.iot-dsa.org*
CitySim	Smart traffic and smart city simulation, 3D visualisation, large scale smart city automation; *http://www.citisim.org/*
Node-RED	Flow based programming; *http://www.nodered.org*
IoTivity	On-board Constrained Application Protocol (CoAP) as application layer; *https://www.iotivity.org*
KAA	Data analytics, dynamic updates in real-time; *https://www.kaaproject.org*

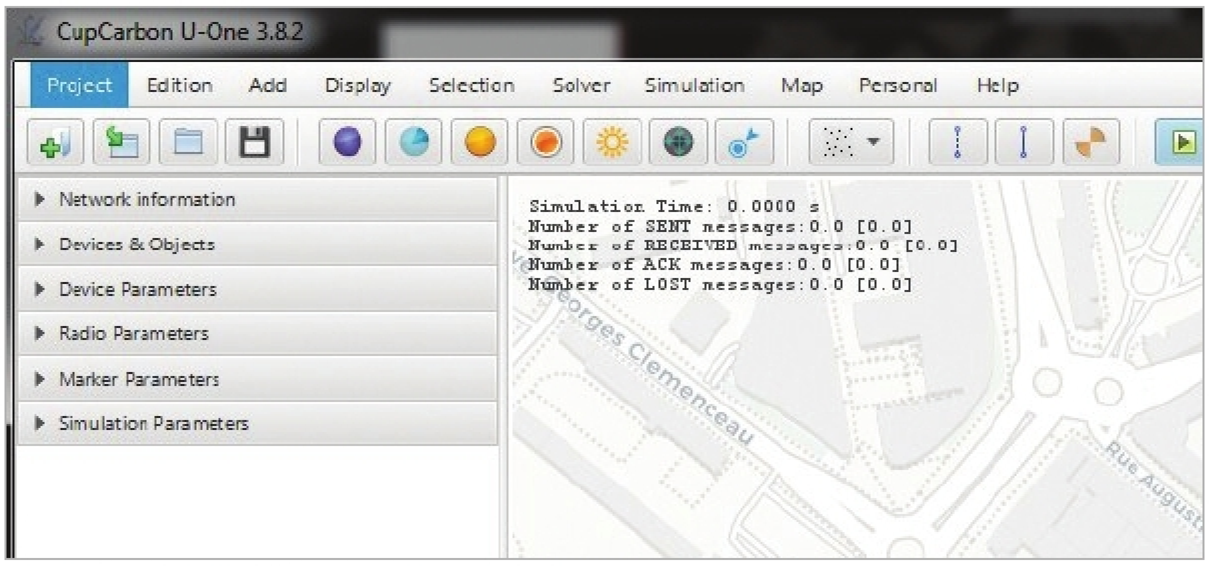

Figure 4: Working panel of the CupCarbon simulator

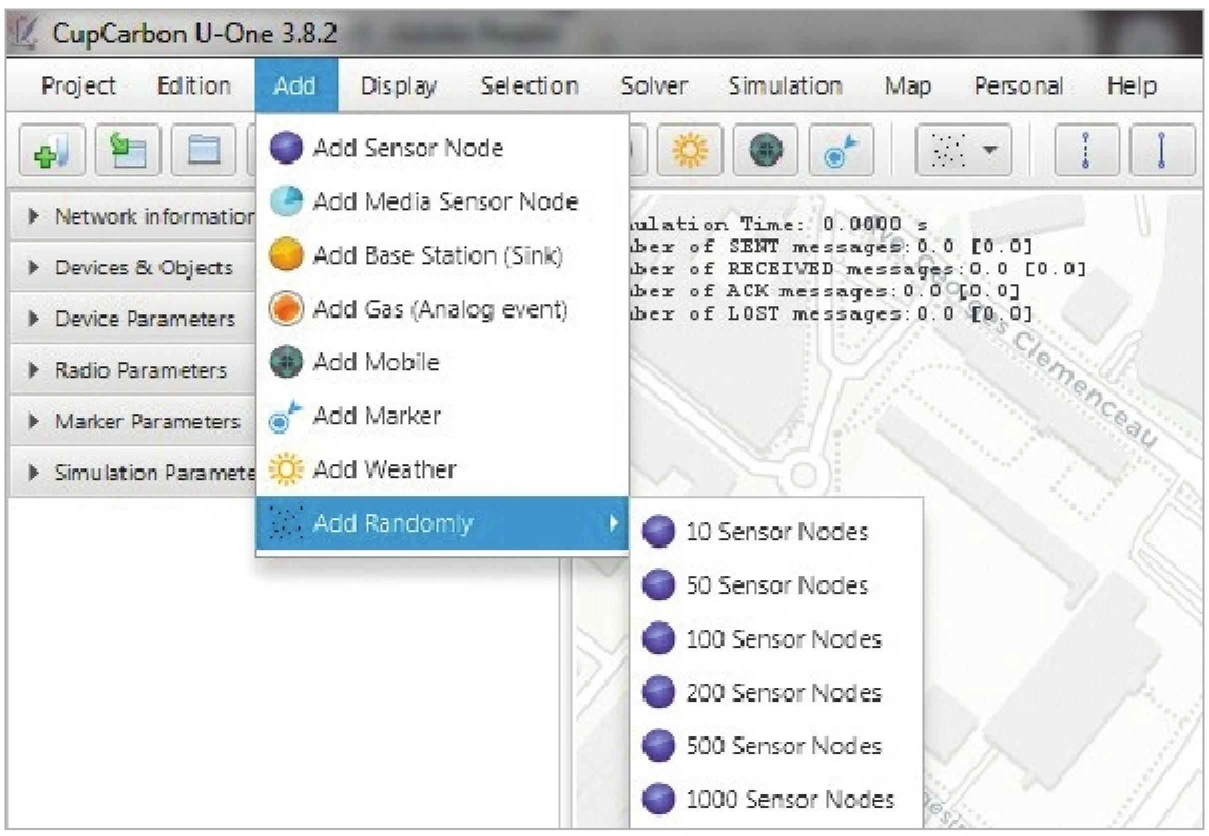

Figure 5: Adding different types of sensor nodes in CupCarbon

standard with a low-power wide-area network (LPWAN) that enables wide scale coverage and better performance of connected smart devices.

Free and open source tools for IoT implementation

A wide range of free and open source simulators and frameworks is available to simulate IoT scenarios. These can be used for R&D so that the performance of different smart city and IoT algorithms can be analysed. Research projects for a smart city need to be simulated so that the citizen behaviour can be evaluated on multiple parameters before launching the actual IoT enabled smart city systems.

Installing and working with CupCarbon

CupCarbon (*http://www.cupcarbon.com*) is a prominent, multi-featured simulator that is used for the simulation of smart cities and IoT based advanced wireless network scenarios.

It provides an effective graphical user interface (GUI) for the integration of objects in the smart city with wireless sensors. The sensor nodes and algorithms can be programmed in the SenScript Editor in CupCarbon. SenScript is the script that is used for the programming and control of sensors used in the